

Threat Area for Ammonia leak from Killarney Community Center on June 2nd, 2022



Legend



Origin of Ammonia Leak

Ammonia Threat Area



Wind Direction Confidence Lines
(30 ppm/60 mins)



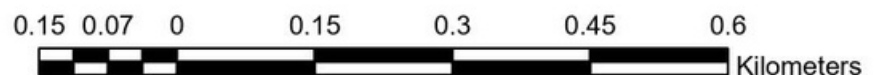
30 ppm (60 mins)



160 ppm (60 mins)



1100 ppm (60 mins)

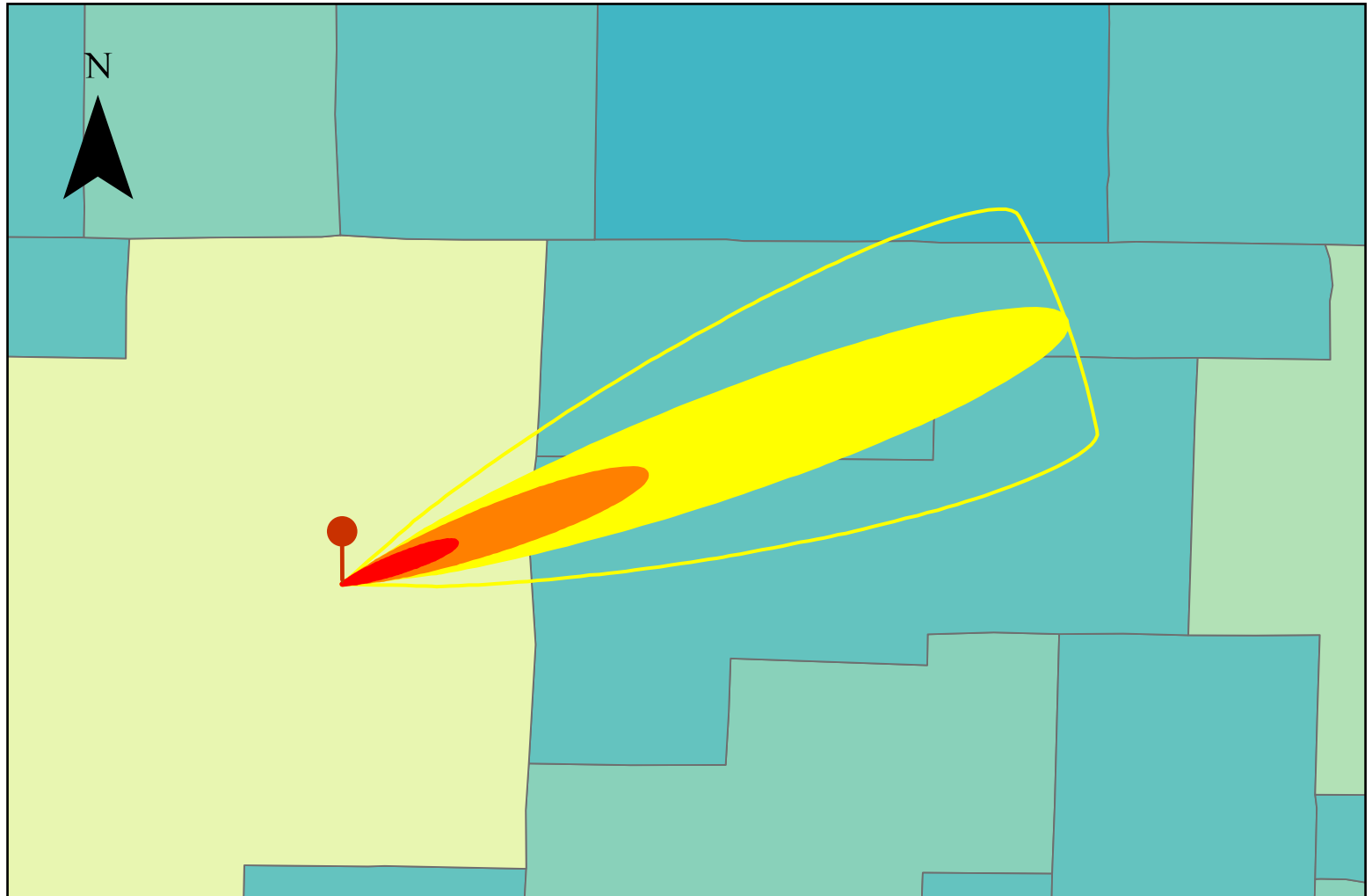


Grady Zappone
20 November 2023

Data Sources: Environment Canada & Census Canada
Coordinate System: WGS 1984

A map depicting the threat area from a potential ammonia leak out of a community center in Vancouver, BC. The leak occurs at a rate of 2.77 pounds per second for a period of 1 hour. The area is overlaid above satellite imagery.

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Origin of Ammonia Leak

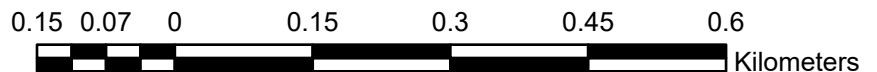
Vancouver
Dissemination
Areas

Population Density
(people per square
kilometer)

0.000000 - 837.982833
837.982834 - 1883.064516
1883.064517 - 2804.232804
2804.232805 - 3697.339246
3697.339247 - 4777.534646
4777.534647 - 6038.461538
6038.461539 - 7542.764051
7542.764052 - 9470.365700

Ammonia Threat
Area

- Wind Direction
Confidence Lines (30
ppm/60 mins)
- 30 ppm (60 mins)
- 160 ppm (60 mins)
- 1100 ppm (60 mins)

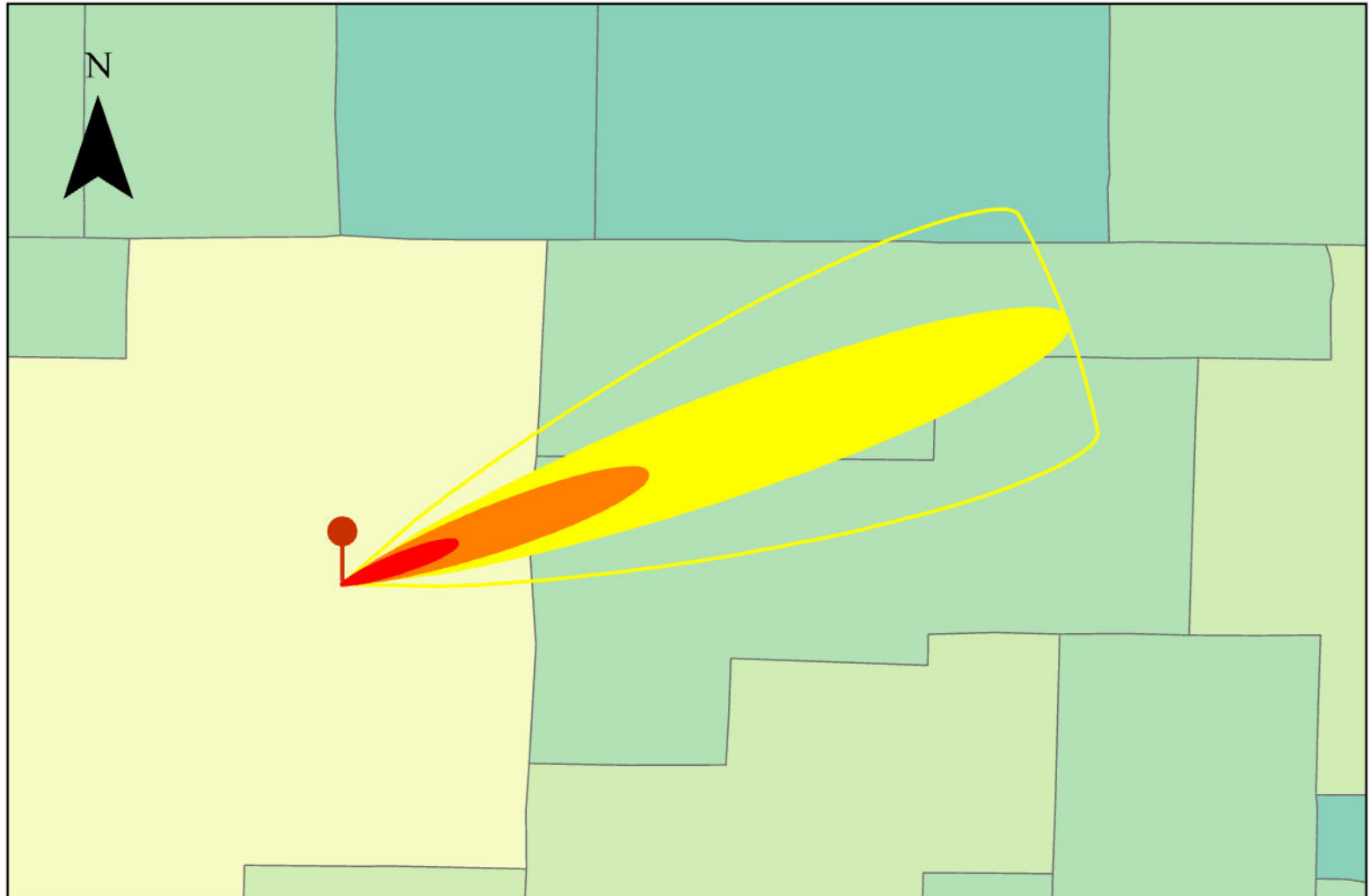


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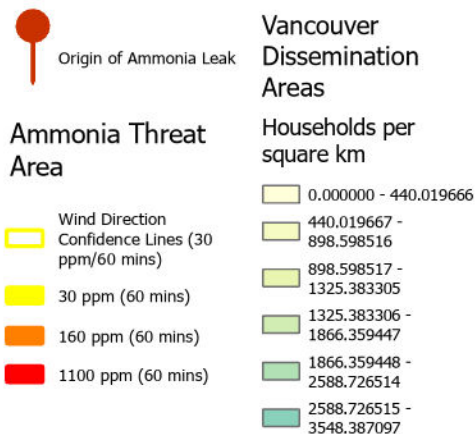
Data Sources: Environment Canada & Census Canada
Coordinate System: WGS 1984

A map depicting the threat area from a potential ammonia leak out of a community center in Vancouver, BC. The leak occurs at a rate of 2.77 pounds per second for a period of 1 hour. The threat area is overlaid over Vancouver dissemination areas, displaying the population density of each area.

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Data Sources: Environment Canada & Census Canada
Coordinate System: WGS 1984

A map depicting the threat area from a potential ammonia leak out of a community center in Vancouver, BC. The leak occurs at a rate of 2.77 pounds per second for a period of 1 hour. The threat area is overlaid over Vancouver dissemination areas, displaying the household density of each area.

The analysis of the potential impact of a worst-case scenario involving an ammonia leak from a community center in Vancouver found that concentrations in the threat zone are estimated at 30 ppm or higher. The Apportion Polygon tool identified that the zone encompasses approximately **203 households**, with an estimated **620 individuals** potentially affected. These figures are based on census data aggregated at the Dissemination Area (DA) level.

DAs are typically delineated based on administrative boundaries, which may not align with natural or human-made features. This arbitrary delineation introduces the Modifiable Areal Unit Problem (MAUP), potentially impacting the accuracy of our estimates. One of the DAs within the threat zone includes a park, a large area without residents. Due to the presence of this empty space, our analysis might overestimate the actual population at risk. This overestimation is because the park increases the overall area of the DA without contributing to its population count, skewing the population density and risk exposure figures. If the DAs had been drawn a different way, the estimated number of people and households would have been different.